ACADEMIC CENTRAL

PRACTICE EXAMINATION

# NC(V) LEVEL 2

# MECHATRONICS

## PAPER 1 · PRACTICE C

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Paper 1 style (theory). Section A: objective questions (30) · Section B: mechatronics, PPE and tools, levers and pulleys with moments, mechanical advantage and efficiency, gears, belt drives and mechanisms, materials, relays and pneumatic cylinders (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In a first-class lever the fulcrum is …

A between the effort and the load. B at one end, with the load in the middle.
C at one end, with the effort in the middle. D outside the lever.

1.1.2 Mechanical advantage is calculated as …

A effort ÷ load. B load ÷ effort.
C distance × load. D effort × load.

1.1.3 A wheelbarrow is a … lever.

A first-class B third-class C second-class D fourth-class

1.1.4 The tool used to measure a small diameter to 0,01 mm is a …

A tape measure. B steel rule. C micrometer. D protractor.

1.1.5 The best electrical insulator of the following is …

A copper. B PVC. C aluminium. D steel.

1.1.6 Pneumatic systems work with …

A compressed air. B oil. C water. D steam.

1.1.7 The SI unit of force is the …

A newton. B joule. C watt. D pascal.

1.1.8 A pulley system with 4 ropes supporting the load has an ideal mechanical advantage of …

A 2 B 4 C 8 D 1

1.1.9 PPE stands for …

A power plant energy.
B pneumatic pressure equipment.
C portable power electronics.
D personal protective equipment.

1.1.10 The movement of a piston in a cylinder is …

A rotary.
B oscillating.
C circular.
D reciprocating (back and forth in a line).

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A mechatronic system combines mechanics, electronics and computing.

1.2.2 Two gears that mesh turn in the same direction.

1.2.3 Velocity ratio = distance moved by the effort ÷ distance moved by the load.

1.2.4 The efficiency of a real machine can be more than 100%.

1.2.5 Steel is an alloy of iron and carbon.

1.2.6 Bolts and nuts are permanent fasteners.

1.2.7 A relay uses a small current to switch a larger current.

1.2.8 A double-acting cylinder extends and retracts by air pressure.

1.2.9 Loose clothing is safe near rotating machines.

1.2.10 Limit switches detect the position of moving parts.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Pivot point of a lever | A | relay |
| 1.3.2 | Force applied to a machine | B | efficiency |
| 1.3.3 | Gear between the driver and driven gear that keeps their direction the same | C | fulcrum |
| 1.3.4 | Output work divided by input work | D | vernier caliper |
| 1.3.5 | Metal pin used to join sheets permanently | E | effort |
| 1.3.6 | Changes rotary motion into straight-line motion with a toothed bar | F | actuator |
| 1.3.7 | Electromagnetically operated switch | G | idler gear |
| 1.3.8 | Changes energy into movement | H | emergency stop |
| 1.3.9 | Measures lengths to about 0,02 mm | I | rivet |
| 1.3.10 | Stops a machine immediately in danger | J | rack and pinion |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2: MECHATRONICS, SAFETY AND TOOLS

**2.1** Define mechatronics and give TWO examples of mechatronic products. (3)

**2.2** Name FOUR items of PPE and the hazard that EACH protects against. (4)

**2.3** Name the tool used to: (a) cut thin wire; (b) tighten a hexagonal nut; (c) measure a shaft diameter to 0,01 mm; (d) mark the centre of a hole before drilling. (4)

**2.4** Give THREE safety rules for working with machines in a workshop. (3)

**[14]**

## QUESTION 3: LEVERS AND PULLEYS

**3.1** An effort of 50 N is applied 1,2 m from the fulcrum of a lever. The load is 0,3 m from the fulcrum on the other side. Calculate:

3.1.1 The load that can be lifted (use moments) (3)

3.1.2 The mechanical advantage (1)

3.1.3 The velocity ratio (1)

**3.2** A pulley system lifts a load of 600 N with an effort of 200 N. The effort moves 4 m while the load rises 1 m. Calculate the mechanical advantage, the velocity ratio and the efficiency. (4)

**3.3** Name the THREE classes of levers and give ONE example of EACH. (6)

**3.4** Give ONE reason why the efficiency of a real machine is always less than 100%, and TWO ways to improve it. (3)

**3.5** Calculate the work done to lift the 600 N load through 1 m. (2)

**[20]**

## QUESTION 4: GEARS, DRIVES AND MECHANISMS

**4.1** A driver gear with 15 teeth turns at 900 r/min and drives a gear with 45 teeth. Calculate the gear ratio and the speed of the driven gear. (3)

**4.2** A 20-tooth idler gear is placed between the two gears. What effect does it have on the gear ratio and on the direction of the driven gear? (2)

**4.3** An 80 mm pulley on a motor turning at 1 200 r/min drives a 240 mm pulley. Calculate the speed of the driven pulley. (2)

**4.4** Give TWO advantages of a gear drive and TWO advantages of a belt drive. (4)

**4.5** Name FOUR mechanisms that change one type of motion into another and give ONE use of EACH. (4)

**4.6** The input torque to the gears in Question 4.1 is 2 N·m. Calculate the output torque, ignoring losses. (2)

**4.7** Why must gears be lubricated? (1)

**[18]**

## QUESTION 5: MATERIALS, ELECTRICAL AND PNEUMATIC SYSTEMS

**5.1** Choose a suitable material for EACH part and give a reason: (a) a light robot frame; (b) electrical wiring; (c) small gears in a robot toy; (d) cable insulation. (8)

**5.2** Explain how a relay allows an Arduino to switch on a 12 V motor. (3)

**5.3** Explain the difference between a single-acting and a double-acting pneumatic cylinder. (4)

**5.4** Name THREE main parts of a basic pneumatic system. (3)

**[18]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**